Lab2: Git Private Repo Integration with Jenkins (Docker)

Goal

This lab deals with both Jenkins configuration and private Github repository. The goal of this lab is to learn how to configure Jenkins environmental variable and create ssh private/public key and authenticate with Github to checkout its private repo.

To check out code from a **private** GitHub repository using the official `jenkins/jenkins:lts` Docker image, you need to spin up the container and provide Jenkins with authenticating credentials. The image already comes with `git` pre-installed, so you do not need to modify the image itself. (ref1, ref2)

You can accomplish this using either **HTTPS (Personal Access Token)** or **SSH (Deploy Keys)**. (ref1, ref2)

Knowledge

- Private/Public key

You can find some introduction here,

- https://kevinli-webbertech.github.io/blog/html/courses/cybersecurity/public_private_key.html
- https://kevinli-webbertech.github.io/blog/html/courses/cybersecurity/mathematics.html
- Jenkins Environmental variables
- Github Configurations

Labs

Step 1: Run the `jenkins/jenkins:lts` Container

Start by running the official LTS container: (ref1)

```
docker run -d \
 --name jenkins \
 -p 8080:8080 \
 -p 50000:50000 \
 -v jenkins_home:/var/jenkins_home \
 jenkins/jenkins:lts
```

Retrieve your initial administrator password by running `docker logs jenkins`, navigate to `http://localhost:8080`, and finish the initial setup wizard by installing the recommended plugins (which includes the **Git plugin**). (ref1, ref2)

Step 2: Choose Your Authentication Method

Option A: Authenticate via HTTPS (Personal Access Token) - Easiest

1. **Generate a PAT on GitHub**: Go to **Settings** > **Developer Settings** > **Personal Access Tokens** > **Tokens (classic)**. Click **Generate new token** and check the `repo` scope box. Copy the generated token. (ref1, ref2)
2. **Add Credentials to Jenkins**: Go to **Manage Jenkins** > **Credentials** > **System** > **Global credentials** > **Add Credentials**.
3. **Kind:** Username with password
4. **Username:** Your GitHub username
5. **Password:** Paste your GitHub Personal Access Token
6. **ID:** `github-token` (or a descriptive name) (ref1, ref2)
7. **Configure your Pipeline/Job**: Set the Repository URL to the HTTPS variant (e.g., `https://github.com/<username>/<repo>.git`) and select your newly created credential from the dropdown menu. (ref1)

Option B: Authenticate via SSH (Deploy Keys) - More Secure

1. **Generate SSH Keypair**: Open your terminal and generate an SSH key pair: (ref1)

`bash ssh-keygen -t ed25519 -C "jenkins-deploy-key" -f ./jenkins_key`

1. **Add Public Key to GitHub**: Go to your GitHub Private Repository > **Settings** > **Deploy keys** > **Add deploy key**. Paste the contents of `jenkins_key.pub`. (ref1)
2. **Add Private Key to Jenkins**: Go to **Manage Jenkins** > **Credentials** > **System** > **Global credentials** > **Add Credentials**.
3. **Kind:** SSH Username with private key
4. **Username:** `git`
5. **Private Key:** Select **Enter directly** and paste the entire contents of your private key file (`jenkins_key`).
6. **ID:** `github-ssh` (ref1, ref2)
7. **Configure your Pipeline/Job**: Use the SSH repository URL (e.g., `git@github.com:your-username/your-repo.git`) and link your SSH credentials. (ref1)

Step 3: Implement Checkout in your Jenkinsfile

Once your credentials are configured, you can call them seamlessly inside a Jenkins Pipeline script using the `checkout` or `git` syntax: (ref1)

```
pipeline {
   agent any
   stages {
       stage('Checkout Code') {
           steps {
               // If using Option A (HTTPS)
               git url: 'https://github.com/<username>/<repo>.git',
                   branch: 'main',
```

```
                    credentialsId: 'github-token'

                // OR If using Option B (SSH)
                // git url: 'git@github.com:your-username/your-repo.git',
                //    branch: 'main',
                //    credentialsId: 'github-ssh'
            }
        }
    }
}
```

Troubleshooting

If you run into issues, check the following:

- Which **authentication type** (HTTPS token or SSH key) you are using.
- Any **error codes** or console output you see during checkout.

Ref

- https://piotrminkowski.com/2018/06/12/local-continuous-delivery-environment-with-docker-and-jenkins/
- https://www.youtube.com/watch?v=vsEI0HIv8b4
- https://medium.com/@mai.vly/setup-jenkins-for-your-application-from-the-private-github-repo-with-docker-bac793
- https://tesh.digital/set-up-a-cicd-server-and-configure-for-use-with-private-github-repositories
- https://www.youtube.com/watch?v=Xs6bXpJsGu4&t=150
- https://www.youtube.com/watch?v=t-UAzkZTKZI&t=362
- https://stackoverflow.com/questions/9767919/in-jenkins-how-to-checkout-a-project-into-a-specific-directory-using-

Lab Document

- https://kevinli-webbertech.github.io/blog/md/courses/dev_ops/github/labs/GitPrivateRepoIntegrationWithJenkin.pd